

Research Problem Identification and Formulation

Identification of Research Problem

Formulation of problem is the first and foremost step in a research work. The research problem can be formulated and selected rationally and then the whole research work can be conducted only if the identification of the problem is done precisely. To identify the problem a researcher should have some basic knowledge, which is then developed through discussion with experts or through the literature or by continuation of activities in the related field.

The research problem undertaken for the study must be carefully selected. This task is a difficult one, although it may not appear to be so.. Help may be taken from a supervisor in this connection. A research supervisor can at the most only help a researcher to choose a subject. Identifying the exact nature and dimensions of a problem is of major importance in research work. It is very essential that an investigator should learn how to recognize and define a problem.

The following steps are to be followed in identifying a research problem:

1. Determining the field of research in which a researcher is keen to do the research work.
2. Develop the mastery on the area or in the field of specialization.
3. Review the recent research conducted in the selected area.
4. On the basis of review, select the priority field of the study.
5. Draw an analogy and insight in identifying a problem or employ the personal experience of the field in locating the problem. In this process researcher can take help of supervisor or expertise of the field.
6. Pin-point specific aspect of the problem which is to be investigated.

Ways of Understanding Research Problem

The selection of a suitable problem is not an easy task. It is a serious responsibility to commit oneself to a problem that will inevitably require much time and energy and which is so academically significant. Specifically, the concept for separating out the research problem from the diversified field can be made by the

- i. Discussion among the colleagues
- ii. Discussion with the research guide
- iii. Discussion with some experts and
- iv. Intensive reading all the available literature.

The following are the general ways for understanding problem to which one may proceed for a suitable research:

1. Personal experience of the investigator in any field is the main means for understanding a suitable problem.
2. The other ways of understanding of problem, most frequently used by the investigator as suggested by the supervisors, is the extensive study of available literature-research abstracts, journals, hand-books of research

international abstracts etc

3. In the choice of a suitable problem, the researcher has to decide his field of investigation. He should study the field intensively in the specific area; this may enable him to identify a problem from the specific field.
4. The new innovations, technological changes and curricular developments are constantly bringing new problems and new-opportunities for social research.
5. The most practical ways of understanding problem is to consult supervisor, experts of the field and most experienced person of the field
6. It is a general practice that researchers suggest some problems in their research reports. The researcher can understand a suitable problem for his own study.

Steps in Research Problem Formulation

In a scientific inquiry, the formulation of general topic into a specific problem is the formulation of research problem. The researcher must try to understand the whole problem thoroughly and then rephrasing the same into meaningful terms from an analytical point of view. The following points are suggested as the steps for the formulation of research problem.

- Statement of problem in general way
- Understanding the nature of the problem
- Surveying the available literature
- Developing the ideas through discussion
- Rephrasing the research problem.

Criteria of a Good Problem

Fred. N. Kerlinger has defined a good problem as an inquisitive sentence statement that asks what relation exists between two or more variables. The research questions, research objectives and the hypotheses of the research lie on the problem statement of the research work. So for selecting research questions, setting research objectives and the hypothesis of the research selected problem should be an ideal one. An ideal research problem must have the following three main criteria:

- 1) *The problem should express a relation between two or more variables.* The topic of the research work reflects the research problem and it would be inquisitive and interesting if stated in terms of relation between variables.

- 2) *The problem should be stated rigidly and unambiguously.* If a research problem has ambiguity in its interpretation consequently it affects research design and the whole research process and hence the result of the study.
- 3) *The problem statement should be such as to imply possibilities of empirical testing.* In case, when problem statement expresses the relationship between two variables and is stated rigidly but if the testing the relationship in the circumstances of adopted research design and within the available facility is impossible then the selected problem cannot be good problem. Thus, a good research problem should always be linked with the research design and it must consider the available facility and capability of the study.